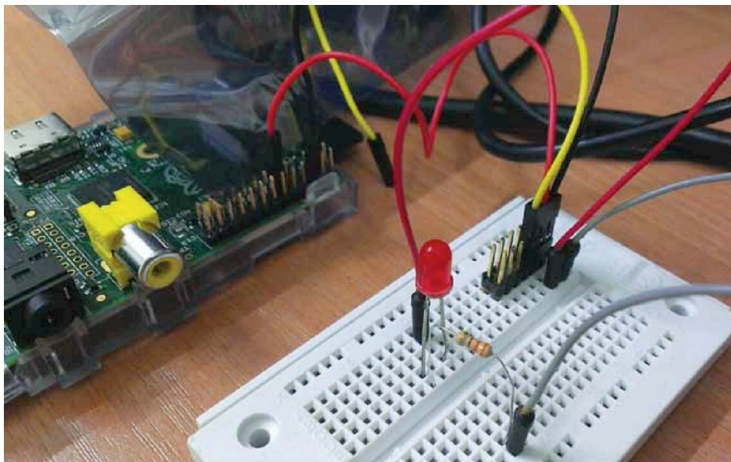


Compile a GPIO Control Application and Test It On the Raspberry Pi

GPIO is the acronym for General Purpose (I/O). The role played by these drivers is to handle I/O requests to read or write to groups of GPIO pins. Let's try and compile a GPIO driver.



This article goes deep into what really goes on inside an OS while managing and controlling the hardware.

The OS hides all the complexities, carries out all the operations and gives end users their requirements through the UI (User Interface). GPIO can be considered as the simplest of all the peripherals to work on any board. A small GPIO driver would be the best medium to explain what goes on under the hood.

A good embedded systems engineer should, at the very least, be well versed in the C language. Even if the following demonstration can't be replicated (due to the unavailability of hardware or software resources), a careful read through this article will give readers an idea of the underlying processes.

Prerequisites to perform this experiment

- C language (high priority)
- Raspberry Pi board (any model)
- BCM2835-ARM-peripherals datasheet (just Google for it!)

- Jumper (female-to-female)
- SD card (with bootable Raspbian image)

Here's a quick overview of what device drivers are. As the name suggests, they are pieces of code that drive your device. One can even consider them a part of the OS (in this case, Linux) or a mediator between your hardware and UI.

A basic understanding of how device drivers actually work is required; so do learn more about that in case you need to. Let's move forward to the GPIO driver assuming that one knows the basics of device drivers (like inserting/removing the driver from the kernel, probe functionality, etc).

When you insert (*insmod*) this driver, it will register itself as a platform driver with the OS. The platform device is also registered in the same driver. Contrary to this, registering the platform device in the board file is a good practice. A peripheral can be termed a platform device if it is a part of the SoC (system on chip). Once the driver is inserted, the registration (platform device and platform driver) takes place,